developed which can be guided to repair such items as unshielded nuclear reactors, radionuclide equipment, heavily contaminated radiochemical apparatus, etc.

15.15. CONTROL OF RADIATION PROTECTION MEASURES

In larger organizations, the control of radiation hazards is the responsibility of specialists known as *health physicists* or, sometimes, *health chemists*. Their main duty is to ensure that work is carried out without hazard to the health of the people involved.

The protection follows three stages: prevention, supervision, and after-control. Preventive measures include use of fume hoods, α-boxes, radiation shielding, tongs, etc., as discussed above. The supervision stage involves the use of radiation instruments to monitor the radiation level (see Ch. 9). Small TLD, film or pocket pen dosimeters are used for individual monitoring (§8.9). For spills and contamination of hands, shoes, etc., special contamination instruments (counters) are used which are more sensitive than the monitoring dose instruments.

Contamination in the laboratory must be avoided. This is controlled by smear tests; i.e., a filter paper is wiped over the surface and the paper is checked with a suitable instrument. In a so-called "clean area", the fixed contamination should not exceed 2 Bq for α, and 4 Bq for β-γ on a surface of 100 cm^2. For an "active area" the rules are a maximum of 20 Bq for α per 100 cm^2, and 0.01 mGy h^{-1} from β—γ at a distance of 2 cm from the surface. Radioactive aerosols are monitored by air samplers in which a certain amount of air is drawn through a fine filter paper after which the paper activity is measured.

The after-control usually consists of checking personal dosimeters and a medical examination. Depending on the kind of work, the dosimeters are checked from twice a week to once a month. A medical examination is given once or several times a year, depending on the work conditions. In danger of inhalation or ingestion of α- or soft β-emitters, urine samples are analyzed. Such analyses are very sensitive and much less than a kBq in the body is easily detected. For workers who handle hazardous amounts of α-emitters (e.g. plutonium in more than milligram amounts), urine samples should be taken regularly.

If necessary whole-body counts are also taken. *Whole-body counting* is carried out with the subject being surrounded by numerous scintillation or solid state detectors in a heavily shielded room. The natural body content of ^{40}K is easily detected and is a control of the efficiency of this technique.

15.16. EXERCISES

15.1. "Reference man" consists of 18%C, 66% H_2O, 0.2% of K per body weight. He may also have accumulated 10 pCi ^{226}Ra in the body; assume 0.3 decay for each of

the following 5 daughters. Calculate for a body weight of 70 kg the number of radioactive decays per unit time from ^{3}H, ^{14}C, ^{40}K and ^{226}Ra. Assume 30 TU in water.

15.2. Using the information above, how many grams of the body's molecules (assume average mole weight of 10^5) will be damaged in a year if the G(damage) value is 3.1×10^{-7} mol/J? Assume $E_{abs}(\beta) = E_{max}/3$.

15.3. Under the same assumption as above, what amount of damage will be caused by cosmic radiation? Assume that the cosmic particles produce 3×10^9 ion pairs s^{-1} m^{-3} of the body.

15.4. With the information in exercise 15.1, calculate annual doses received from (a) ^{40}K, and (b) ^{226}Ra and daughters. Assume $w_r(\alpha) = 20$.

15.5. Ten mg ^{238}U has been collected in the kidneys. Considering the biological half-life of uranium and assuming only one α-emission in ^{238}U decay, calculate the dose (in Sv) received by the organ if the uranium is evenly distributed. The weight of a kidney is 150 g.

15.6. A γ-dose rate of 1 Sv is assumed to inactivate (kill) human cells. The body contains 6×10^{13} cells in a cell weight of 42 kg for a 70 kg man. (a) What average energy (in eV) has to be deposited in a cell to kill it? (b) Calculate the number or kidney cells destroyed for the dose received in exercise 15.5. For simplicity assume the cells to be cubic with a side length of about 11 μm.

15.7. A tumor has the weight of 80 g and we wish to destroy 20% of the cells by irradiating with 180 MeV protons with such penetration that half of the energy is deposited in the tumor. The particle beam is 5 μA. For what time must the irradiation be? A cell of weight 10^{-9} g is assumed to be killed on the absorption of 200 keV and no cell is assumed to be killed twice.

15.17. LITERATURE

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